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RESEARCH ARTICLE

AGM-YOLO: A Generalizable Plug-and-Play Attention Module for Enhancing Small Object Detection Across Biomedical and Remote Sensing Imagery

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ABSTRACT The accurate detection of minute objects in biomedical images is a critical bottleneck for early disease diagnosis. These targets pose a significant challenge to existing algorithms because of their small size, low signal-to-noise ratio, and tendency to be obscured by complex backgrounds. Although general-purpose detection methods have advanced considerably, they still exhibit deficiencies in precision and recall for such diminutive targets. To address this, we designed and proposed a novel plug-and-play neural network module, termed AGM-YOLO, engineered to significantly enhance the detection capabilities of YOLO-family architectures. The core innovation of AGM-YOLO lies in its sophisticated synergistic multi-attention architecture, which combines a customized multi-head self-attention (CMHA), a lightweight channel attention block (GEBlock), and an efficient multiscale feature refinement neck (MFR-Neck). This design systematically increases the detection performance in multiple dimensions. Extensive experiments on three public datasets spanning the biomedical and remote sensing domains demonstrated that YOLO models equipped with AGM-YOLO substantially outperformed various state-of-the-art baselines. By setting a new benchmark for both precision and recall, AGM-YOLO shows considerable potential for advanced automated clinical diagnosis.

INDEX TERMS Minute object detection, biomedical imaging, attention mechanism, feature fusion, YOLO, AGM-YOLO.

I. INTRODUCTION

Biomedical image analysis serves as a cornerstone of modern clinical practice [1] and plays a pivotal role in precise diagnosis, personalized treatment planning, and assessment of therapeutic efficacy [2]. The rapid advancement of high-throughput imaging technologies has generated an unprecedented volume of medical data, creating immense opportunities for automated computer vision-based disease screening. Among these applications, the accurate detection of minute lesions at the earliest stages is critical for improving the prognosis of major diseases. Our work, while not a standalone diagnostic tool, serves as a crucial enabling technology for computer-aided diagnosis (CAD) systems. As demonstrated on authoritative benchmarks, such as the LUNA16 dataset, its primary application lies in assisting clinicians in the early detection of potential pulmonary abnormalities,such as cancerous nodules, a critical first step for the screening and subsequent diagnosis of diseases, such as lung cancer. However, the inherent challenges posed by these targets, namely their diminutive size, ambiguous boundaries, low signal-to-noise ratio, and frequent entanglement with complex anatomical backgrounds, create significant performance bottlenecks. Consequently, existing general-purpose object detection algorithms often suffer from insufficient precision and high miss rates when directly

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applied to this domain, substantially limiting their potential for early clinical diagnosis.

To overcome this bottleneck, attention mechanisms have emerged as a core technology for enhancing the performance of deep-learning models by accentuating salient features while suppressing irrelevant information. Prior research has primarily explored two families of attention: channel attention, which excels in capturing interchannel dependencies, and self-attention, which adeptly models long-range spatial relationships [3]. Despite their merits, these methods have inherent limitations. For example, the quadratic computational complexity of self-attention becomes prohibitive when processing high-resolution medical images. Conversely, many channel attention mechanisms tend to overlook precise spatial information that is vital for localizing minute targets. Therefore, designing an efficient framework that synergistically harnesses the strengths of different attention mechanisms while mitigating their respective drawbacks remains a pivotal and unresolved question in the field of biomedical object detection.

To address this challenge, this paper presents an innovative solution based on the efficient You Only Look Once (YOLO) architecture. We designed and implemented a plug-and-play attention module, termed AGM-YOLO, which was engineered to systematically improve the accuracy and robustness of minute object detection. The central contribution of this work lies in the novel synergistic architecture co-designed within the AGM-YOLO module. It organically integrates a customized, computationally efficient multi-head self-attention (CMHA) mechanism with a lightweight channel attention block (GEBlock) and efficient multiscale feature refinement neck (MFR-Neck). This synergistic design enables the model to concurrently capture both the global context and fine-grained local details while adaptively enhancing the most discriminative information to precisely focus on minute targets. To validate the efficacy and generalizability of our approach, we conducted comprehensive experiments on three public datasets spanning both biomedical imaging and remote sensing. The results demonstrate that models equipped with AGM-YOLO significantly outperform various state-of-the-art methods, not only providing a more reliable tool for biomedical image analysis but also proving its versatility and substantial potential in cross-domain small object detection tasks. Finally, through comprehensive experiments on datasets from two distinct domains, biomedical and optical remote sensing, we demonstrated the effectiveness and superior generalizability of our proposed method.

While prior works have explored channel attention (e.g., SE-Net, CBAM) or self-attention (e.g., Vision Transformer) in isolation, the core innovation of AGM-YOLO lies in its synergistic co-design within a single, unified architecture. This organic integration of a convolution-based self-attention for the global context, lightweight channel attention for feature refinement, and efficient multi-scale fusion neck differentiates our approach. This holistic design philosophy allows our module to address these multifaceted challenges simultaneously and efficiently, leading to a more robust and generalizable solution.

II. RELATED WORK

Small Object Detection (SOD) remains a formidable challenge in computer vision. To provide a quantitative basis for our work, we adhere to the standard definition used in the COCO benchmark, where 'small objects' are defined as objects with a bounding box area smaller than 32×32 pixels. These targets pose significant challenges to standard detection frameworks because of their limited pixel information and susceptibility to being overwhelmed by background noise [4]. This challenge is particularly acute in biomedical image analysis, where the precise identification of early stage minute lesions is directly linked to disease diagnosis and patient prognosis [5]. Despite the tremendous success of deep learning-based object detectors, their design paradigms are often skewed towards regular-sized objects, leading to significant performance degradation when detecting minute targets [6]. To this end, the research community has explored various avenues to enhance SOD performance. This section reviews three key research directions that are most relevant to our proposed AGM-YOLO module: advanced attention mechanisms, multiscale feature fusion strategies, and mainstream YOLO detection framework.

A. ATTENTION MECHANISMS IN OBJECT DETECTION

Mimicking the human visual system, attention mechanisms have become an indispensable component of modern deep learning models by enabling them to adaptively focus on the most critical parts of the input. In object detection, they help the model precisely identify targets within cluttered backgrounds.

Attention mechanisms can be broadly categorized as follows. Channel attention, pioneered by the Squeeze-and-Excitation Network (SE-Net) [7], explicitly models interchannel dependencies, allowing the network to learn which feature channels to accentuate. Subsequent works like CBAM [8] the Convolutional Block Attention Module (CBAM) combined channel attention with spatial attention to further refine feature representation. Gather-and-Excite Block (GEBlock) [9], an efficient and lightweight module, leverages a large receptive field to aggregate the spatial context that guides the generation of channel weights, which is beneficial for enhancing weak target signals [10]. However, such mechanisms, which focus primarily on local or channel-wise information, may falter in highly complex scenes where a global contextual understanding is required.

Another major category is self-attention, epitomized by the Transformer [11] and its visual counterpart, the Vision Transformer (ViT) [12]. By computing the relationships between all pairs of pixels, self-attention excels at capturing long-range global dependencies. However, its $O(n^2)$ computational complexity is often prohibitive for high-resolution medical images, and it typically requires extensive pre-training

data. Therefore, efficient and complementary fusion of the efficiency of lightweight attention with the global modeling capability of self-attention (or its variants) is a critical pursuit for enhancing SOD. This directly motivates the multi-attention co-design philosophy within the AGM-YOLO.

B. EVOLUTION OF MULTISCALE FEATURE FUSION STRATEGIES

The hierarchical nature of CNNs naturally produces multi-scale features; shallow layers preserve high-resolution spatial details, whereas deep layers contain rich, abstract semantic information. For minute objects, their sparse features are easily lost during successive downsampling. The Feature Pyramid Network (FPN) [13] is a seminal work designed to address this issue. By constructing a top-down pathway with lateral connections, the FPN combines high-level semantics with low-level details, providing high-quality fused features for detection heads at different scales and becoming a standard in modern detectors.

Building on the FPN, researchers have proposed numerous enhancements. PANet (Path Aggregation Network) [14] augmented FPN with an additional bottom-up path, facilitating the flow of information, particularly strong localization signals from lower layers, to the entire feature pyramid. More recently, BiFPN (Bidirectional FPN) [15], a core component of EfficientDet, introduced a weighted feature fusion mechanism that allows the network to learn the importance of different input features and enables deeper feature fusion through repeated stacking. These advanced designs have significantly improved the detection performance for multiscale objects, particularly small ones. Nevertheless, they can introduce additional computational overhead, and during fusion, the dominant background features may still " appear as faint target features. Consequently, designing a fusion strategy that is both efficient and capable of adaptively strengthening minute object features is another vital direction for improving the detection robustness.

C. THE YOLO FRAMEWORK AND ITS CHALLENGES

The You Only Look Once (YOLO) [16] series, as a paragon of single-stage detectors, is one of the most widely adopted frameworks owing to its excellent balance of speed and accuracy. From its early versions to contemporary YOLOv8, the series has undergone continuous evolution in its backbone (e.g., CSPDarknet), neck design (e.g., PANet-like structures), and training strategies (e.g., anchor-free, Task-Aligned Assigner). YOLOv8, which features a more flexible decoupled head, demonstrated state-of-the-art performance on multiple benchmarks.

However, the design choices that make YOLO efficient also pose inherent challenges for SOD. First, aggressive downsampling of the backbone is the primary cause of information loss for small targets. Second, a minute object may only be assigned to a single or very few positive samples during training, leading to a severe sample imbalance that biases the model towards the background. Furthermore, without specialized mechanisms, weak target features struggle to compete with strong background features during the feature fusion stage, resulting in missed detection. Therefore, enhancing the YOLO framework's capability for feature preservation, extraction, and discrimination of minute objects, without incurring significant computational costs, is key to its application in high-precision domains such as medical imaging.

D. SUMMARY AND MOTIVATION

The preceding analysis revealed a clear gap in the literature when addressing the demanding task of minute biomedical object detection. Individual attention mechanisms are often a trade-off; advanced feature fusion networks, while powerful, may balance effectiveness against efficiency and noise suppression, and mainstream detectors, such as YOLO, still lack intrinsic sensitivity to minute targets.

This leads to the central question motivating our work: can we design a single, lightweight, plug-and-play module that synergistically addresses these three challenges? To this end, we propose the AGM-YOLO. It is innovatively integrated within a unified architecture:

1) GLOBAL CONTEXT-AWARE MODULE

A computationally efficient, customized multi-head self-attention (CMHA) with convolutional inductive bias to capture the global context;

2) LOCAL FEATURE REFINEMENT MODULE

A lightweight channel attention (GEBlock) to refine local features;

3) MULTISCALE ENHANCEMENT MODULE

An efficient adaptive feature fusion strategy to bolster the representation of small targets across scales;

Our goal was to remedy the deficiencies of existing techniques through this organic co-design, thereby providing a superior solution for high-accuracy, high-efficiency biomedical object detection.

III. METHODOLOGY

To overcome the challenges of detecting minute biomedical targets posed by their size, signal-to-noise ratio, and background complexity (as detailed in the Introduction), this study proposes and designs a plug-and-play Attentional Granularity-aware Module (AGM). This module was integrated into an efficient YOLO framework to form the AGM-YOLO model. The core philosophy of the AGM module is to systematically enhance the perception and discrimination capabilities of the network for minute objects through a meticulously designed synergistic architecture. This chapter first elaborates on the design principles of AGM's core component, the Convolutional Multi-Head Self-Attention (CMHA). Subsequently, the overall architecture of the AGM module is presented, illustrating how it organically

synergizes CMHA, a channel attention block (GEBlock), and the proposed Multiscale Feature Refinement Neck. Finally, the integration strategy of the YOLO architecture is described in detail.

A. CONVOLUTIONAL MULTI-HEAD SELF-ATTENTION (CMHA)

Traditional self-attention mechanisms in vision tasks often flatten feature maps into sequences, which can disrupt local spatial structures that are crucial for object localization. To overcome this limitation, we designed the CMHA module, which aims to combine the potent local feature extraction ability of convolutions with the superior long-range dependency modeling capability of Multi-Head Self-Attention (MHSA).

The key innovation of CMHA is the use of parallel, multiscale convolutional operations to generate the Query (Q), Key (K), and Value (V) projections, instead of standard linear projections. Specifically, for an input feature map $\boldsymbol{X} \in \mathbb{R}^{H\times W\times C}$,CMHA first employs two parallel sets of depth-wise separable convolutions [17] with different kernel sizes (e.g., 3×3 and 5×5) to capture the local contextual information at different granularities. This multiscale design is vital for identifying lesions of various shapes and sizes. The output feature maps from the two convolutional paths are concatenated along the channel dimension. Subsequently, three independent 1×1 convolutional layers were applied to generate Q, K, and V matrices. This design ensures that Q, K, and V are imbued with a rich local spatial context before correlation computation.

The calculation for each head in the MHSA follows the standard-scaled dot-product attention:

$$\text{Attention}(Q_i, K_i, V_i) = \text{softmax}\left(\frac{Q_i K_i^T}{\sqrt{d_k}}\right) V_i \tag{1}$$

where Q_i, K_i, V_i are the results of projecting Q, K, and V into the i-th subspace, and d_k is the dimension of the key vectors, used for stabilizing gradients. The outputs of all heads are concatenated and passed through a final linear transformation to produce the output of the CMHA module, F_{CMHA}. This design not only effectively models global contextual relationships, compensating for the limited receptive fields of pure convolutional networks, but also learns diverse feature representations from different subspaces via the multi-head mechanism, enhancing the generalization ability of the model.

B. OVERALL ARCHITECTURE OF THE AGM MODULE

The AGM module was engineered to implement progressive coarse-to-fine and global-to-local feature refinement processes. It is a sophisticated hybrid architecture that organically integrates CMHA, a GEBlock, and a Multiscale Feature Refinement Neck (MFR-Neck) [18]. The overall architecture and data flow are shown in Fig. 1.

The data processing pipeline within the AGM module is as follows:

1) GLOBAL AND LOCAL CONTEXT ENCODING

The input feature map, X_{in}, first passes through a Layer Normalization [19] and is then fed into the CMHA module (as described in Sec. III-A), thereby yielding an initial context-enhanced feature map, F_{CMHA}.

2) CHANNEL-WISE FEATURE REFINEMENT

To highlight the feature channels crucial for minute object detection, we incorporated a lightweight GEBlock. F_{CMHA} is passed to the GEBlock, which aggregates global spatial information via parallel Global Average Pooling (GAP) [20] and Global Max Pooling (GMP) [21] to form two complementary context descriptors. These descriptors are transformed by a shared Multi-Layer Perceptron (MLP) [22] to generate a channel attention weight vector $\mathbf{w} \in \mathbb{R}^C$. This vector is then applied to F_{CMHA} via channel-wise multiplication to produce the refined feature map F_{GE}:

$$F_{GE} = \sigma\left(MLP\left(GAP\left(F_{CMHA}\right)\right) + MLP\left(GMP\left(F_{CMHA}\right)\right)\right) \otimes F_{CMHA} \tag{2}$$

where σ is the Sigmoid function and $\otimes$ denotes channel-wise multiplication. This step adaptively enhances informative channels while suppressing redundant or noisy channels.

3) MULTISCALE FEATURE REFINEMENT AND FUSION

To effectively handle the diversity in lesion sizes, the final stage of the AGM module processes an enhanced feature map. The feature map F_{GE}, refined by channel attention, is fused with the input feature map F_{CMHA} through an element-wise summation (residual connection). This combined feature map, now rich in both global context and channel-wise salience, is then passed to the Multiscale Feature Refinement Neck (MFR-Neck). The MFR-Neck employs a concise yet effective FPN-like structure for feature fusion. Specifically, it includes a top-down pathway that uses upsampling operations (e.g., bilinear interpolation) and 1×1 convolutions to restore the spatial resolution and merge high-level semantic features with lower-level feature maps. This design ensures the effective integration of information from different receptive fields, producing a final output feature map, F_{AGM}, that possesses both rich semantics and fine-grained spatial details, with a particular emphasis on strengthening the representation of minute objects.

C. INTEGRATION STRATEGY INTO THE YOLO ARCHITECTURE

The AGM module, designed as a plug-and-play component, can be flexibly integrated into the neck of mainstream detection frameworks, such as YOLOv5 and YOLOv8. The neck network in YOLO (typically based on PANet) is crucial for multiscale feature fusion. Through systematic

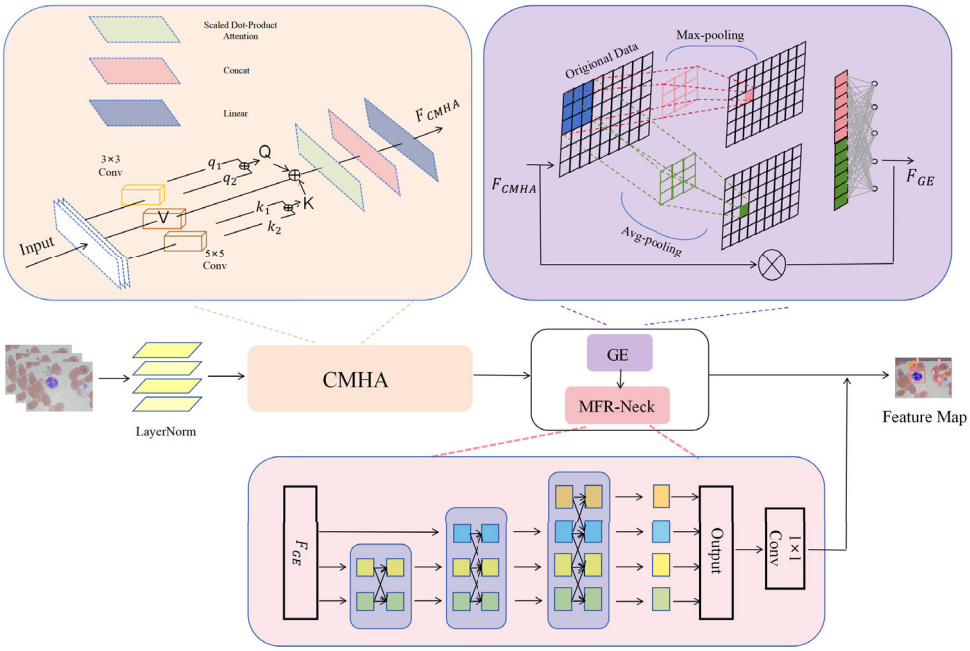

FIGURE 1. Architecture of AGM module.

experimentation, we identified that embedding the AGM module into the processing path of the P4 feature level (at 1/16th of the input resolution) strikes an optimal balance between the performance and efficiency.

The rationale for selecting the P4 level is as follows: This layer acts as a critical pivot connecting deep semantic information (from P5) and shallow detail-rich information (propagating to P3). It retains sufficient spatial resolution to localize small-to-medium objects while being imbued with a rich semantic context. Enhancing features at this level with AGM not only directly improves the detection of medium-sized objects, but more importantly, the enhanced features are subsequently fused with the P3 level features in the top-down pathway. This indirectly boosts the detection capability for minute objects that are primarily handled by the P3 level. This "semantic-guidance, detail-enhancement" synergistic mechanism allows us to achieve significant performance gains, especially for minute objects, by deploying the AGM module at a single location (with <0.5% increase in parameters).

IV. EXPERIMENTS AND ANALYSIS

This chapter aims to rigorously evaluate the effectiveness, generalization capability, and internal component synergy of the proposed AGM-YOLO module through a series of comprehensive experiments. First, the experimental setup, benchmark datasets, and evaluation metrics are introduced. Subsequently, we integrated the AGM module into several mainstream YOLO models, and conducted detailed performance comparisons with their baseline counterparts. Subsequently, we dissected the contribution of each component within the AGM module through ablation studies. Finally, by combining the visualization results with a comprehensive discussion, we provide an in-depth analysis of the experimental phenomena to validate the design principles proposed in our methodology.

A. EXPERIMENTAL ENVIRONMENT AND SETTINGS

To ensure fairness and reproducibility of all experimental results, all models were trained and evaluated on a unified hardware and software platform,The experimental configuration parameters are shown in Table 1. To rigorously test the out-of-the-box generalization capability of our module, the same set of hyperparameters was used across all three datasets without dataset-specific tuning.

To enhance model robustness and prevent overfitting, a comprehensive data augmentation strategy, leveraging the capabilities of the YOLOv8 framework, was employed during training. Geometric transformations included random horizontal flipping (50% probability), scaling (±50%), and translation (±10%). To enrich the learning of small objects and complex backgrounds, Mosaic augmentation was applied with a 100% probability during the initial training phases, a technique known to be highly effective, which was then disabled during the final 10 epochs to stabilize training. Furthermore, color space adjustments were applied, including modifications to HSV channels (hue ±1.5%, saturation ±70%, and value ±40%), to simulate variations in lighting

TABLE 1. Experimental parameter configuration.

Parameters	BCCD	LUNA16	NWPU VHR-10
Hardware Platform	NVIDIA GeForce RTX 3050 Laptop GPU with 4GB VRAM		
Software Framework	PyTorch 2.3		
Image Preprocessing	Resize to 640×640 pixels		
Optimizer	AdamW ($\eta = 1 \times 10^{-3}$, $\lambda = 5 \times 10^{-4}$)		
LR Scheduler	Cosine Annealing		
Epochs	100	300	400
Batch Size	8		
Early Stopping	Stop if mAP@0.5 Plateaus for 10 Epochs		

conditions. This multi-faceted augmentation pipeline ensured the model was exposed to a diverse range of training samples, thereby improving its generalization capabilities across all datasets.

B. EXPERIMENTAL DATASETS

This study selected three public datasets with distinct characteristics and challenges to evaluate the performance and generalization of the proposed method comprehensively. To comprehensively evaluate the performance and, most importantly, the cross-domain generalizability of our proposed AGM-YOLO module, we deliberately selected three public datasets from distinct application domains. These datasets represent three different imaging technologies: microscopy imaging (BCCD), computed tomography (LUNA16), and optical remote sensing (NWPU VHR-10). By validating our method on such diverse data, we aim to demonstrate its robustness and wide applicability.

BCCD (Blood Cell Count and Detection) [23] is a publicly available dataset of microscopic images for blood cell detection, containing 364 images of 640 × 480 pixels (approximately 80MB) with 4,971 annotated objects. These objects are categorized into three classes: red blood cells (RBCs), white blood cells (WBCs), and platelets, with an original class distribution of roughly 20:5:1. The dataset features significant size disparities among objects: RBCs have a diameter of 6-8 pixels, WBCs range from 15-20 pixels, and platelets-representative of "minute objects" in biomedical imaging-are only 2-3 pixels. It also presents challenges such as high RBC overlap (up to 40% in dense regions), low contrast between platelets and plasma backgrounds, and subtle inter-class similarities.

This dataset holds significant value for medical small object detection research: its microscopic imaging simulates clinical environments, minute platelets can serve as ideal surrogates for low-signal small lesions, and dense backgrounds replicate the complexity of biological tissues, making it an excellent benchmark for validating algorithm precision and recall in early disease diagnosis. The dataset was partitioned into training (60%), validation (20%), and testing (20%) sets, with stratification ensuring consistent class and size distributions across the splits.

In thrombocytopenia diagnosis, AGM-YOLO enables accurate counting of 2-3μm platelets, aiding in the identification of immune thrombocytopenic purpura (ITP) and chemotherapy-induced thrombocytopenia, and providing guidance for transfusion decisions and treatment adjustments.

LUNA16 (Lung Nodule Analysis 16) [24] is an authoritative public benchmark dataset in the field of lung nodule detection, derived from clinical chest CT scans. It contains 4,956 slice images with a resolution of 512 × 512 pixels (approximately 12 GB), covering volumetric CT data of 888 patients and 1,186 annotated nodules. The dataset features a diverse distribution of nodule sizes (approximately 35% are micro-nodules <5 mm, 45% are medium-sized nodules 5-10 mm, and 20% are large nodules >10 mm), which is consistent with clinical reality. It also presents challenges such as indistinct nodule boundaries, high similarity to adjacent anatomical structures (prone to false positives), and low signal-to-noise ratio of micro-nodules.

This dataset holds significant value for medical small object detection research: its data is sourced from clinical cases, the characteristics of micro-nodules are consistent with early-stage lung cancer lesions, and it can comprehensively evaluate the robustness of algorithms. Additionally, it is partitioned into training/validation/test sets based on patient cases, ensuring consistent distribution across subsets to enable fair performance comparison between algorithms.

In medical applications, AGM-YOLO can detect sub-centimeter lung nodules (<5mm) in CT scans, supporting the early diagnosis of stage I non-small cell lung cancer (with a 5-year survival rate exceeding 70% after timely surgical resection). It is also sensitive to microcalcifications (0.1-0.5 mm in diameter) in mammograms, aiding in the identification of ductal carcinoma in situ (DCIS), a pre-invasive stage of breast cancer.

NWPU VHR-10 [25] is a high-resolution optical remote sensing dataset, containing 800 images (with a spatial resolution of 0.5-2 meters per pixel, a single image size ranging from 600 × 600 to 4000 × 4000 pixels (averaging approximately 1500 × 1500 pixels), and a total dataset size of about 2.1 GB). It covers 10 object categories including airplanes and ships, with 30,735 annotated objects in total. The dataset features extremely significant object scale variation: small objects occupy only 10-30 pixels, while large objects span 200-500 pixels. It also faces challenges such as cluttered backgrounds (e.g., overlapping buildings and vehicles) and variable illumination (e.g., shadows and cloud cover), and these interferences are similar to "noise" in medical images.

For medical small object detection research, this dataset has unique auxiliary value: although its small objects belong to non-medical categories, they share core characteristics (e.g., low pixel coverage and weak features) with medical minute targets, enabling cross-domain validation of the generalization ability of algorithms. Additionally, the complex background interference in remote sensing images can simulate the anatomical clutter in biomedical images

(e.g., blood vessels around nodules), facilitating the testing of the robustness of background suppression mechanisms, which is crucial for avoiding false positives in medical scenarios. The dataset ensures consistent distribution of object sizes and categories across subsets through stratification, allowing reliable evaluation of model performance.

C. EVALUATION METRICS

We employed standard object detection metrics to quantify model performance: Precision (P), Recall (R), mAP@0.5, and mAP@0.5:0.95. P and R reflect the accuracy and completeness of the model, respectively. mAP@0.5 is the mean Average Precision at an IoU threshold of 0.5, while mAP@0.5:0.95 averages the mAP over multiple IoU thresholds from 0.5 to 0.95, providing a more rigorous evaluation of the model's overall localization and classification performance.

D. COMPARATIVE EXPERIMENTS

To ensure a fair and direct comparison, all experiments were conducted by comparing a baseline model against its AGM-enhanced counterpart on the exact same training, validation, and testing splits. To validate the efficacy of the AGM module, we integrated it into YOLOv5 [26], YOLOv6s, YOLOv8n [27], YOLOv9c [28], YOLOv10n, and YOLO11n, and conducted rigorous comparative tests against their original baseline models on the three aforementioned datasets. Detailed results are presented in Tables 2, 4, and 6.

TABLE 2. Performance comparison between different YOLO baseline models and their variants integrated with the AGM module on the BCCD dataset.

Models	precision	recall	mAP@50	mAP@50-95
YOLOv8n	0.810	0.842	0.874	0.584
Ours(YOLOv8n-AGM)	0.821	0.916	0.894	0.590
YOLOv5	0.843	0.863	0.910	0.612
Ours(YOLOv5-AGM)	0.812	0.905	0.898	0.601
YOLOv6s	0.810	0.906	0.890	0.606
Ours(YOLOv6s-AGM)	0.811	0.910	0.896	0.611
YOLOv10n	0.770	0.804	0.847	0.584
Ours(YOLOv10n-AGM)	0.745	0.753	0.826	0.550
YOLO11n	0.824	0.868	0.886	0.598
Ours(YOLO11n-AGM)	0.849	0.871	0.903	0.610
YOLOv9c	0.820	0.852	0.881	0.573
Ours(YOLOv9c-AGM)	0.884	0.855	0.923	0.631

TABLE 3. Experimental results of detection performance comparison of different models on BCCD.

Models	Precision(%)	Recall(%)	mAP@50(%)
DCBC DeepL [29]	79.90	80.30	82.40
YOLOv5-CW [30]	84.00	85.70	89.90
ADA-YOLO [23]	86.00	85.50	91.20
AGM-YOLO(OURS)	86.40	87.40	91.60

E. ANALYSIS OF EXPERIMENTAL RESULTS

This section provides a detailed analysis of the experimental outcomes. We first discuss the overall efficacy of the AGM module by examining its performance uplift on various YOLO baselines. We then benchmark our best-performing model against other state-of-the-art methods to demonstrate its external competitiveness. Finally, we analyze the nuanced interactions between the AGM module and different baseline architectures.

TABLE 4. Performance comparison between different YOLO baseline models and their variants integrated with the AGM module on the LUNA16 dataset.

Dataset	Nodules	Models	mAP@50	mAP@50-95
LUNA16 Dataset	General Nodules	YOLOv5	0.888	0.468
		Ours(YOLOv5-AGM)	0.915	0.481
		YOLOv10n	0.885	0.463
		Ours(YOLOv10n-AGM)	0.894	0.482
		YOLO11n	0.917	0.489
		Ours(YOLO11n-AGM)	0.924	0.505
	Small Nodules	YOLOv5	0.936	0.566
		Ours(YOLOv5-AGM)	0.938	0.583
		YOLOv10n	0.861	0.507
		Ours(YOLOv10n-AGM)	0.887	0.550
		YOLO11n	0.917	0.556
		Ours(YOLO11n-AGM)	0.930	0.574

TABLE 5. Experimental results of detection performance comparison of different models on LUNA16.

Models	Precision (%)	Recall (%)	mAP@50 (%)	mAP@50-95 (%)	epochs
MS-YOLO [31]	86.7	83.7	88.9	45.8	100
Ghostnetv2 [31]	87.1	84.1	89.2	46.1	100
Gold-YOLO [31]	87.6	85.3	89.5	46.7	100
RT-DETR [31]	88.6	88.2	90.2	47.6	100
CPYOLO [31]	90.8	89.8	92.7	50.6	100
CAF-YOLO [32]	86.9	83.6	90.7	61.7	300
AGM-YOLO (OURS)	91.4	91.7	93.1	51.6	100

1) OVERALL EFFICACY OF THE AGM MODULE

The experimental data presented in our internal comparison experiments (Tables 2, 4, and 6) demonstrates that the proposed AGM module provides a remarkable performance improvement across various YOLO architectures. The module's generalizability is particularly evident on the challenging NWPU VHR-10 remote-sensing dataset (see Table 6). On this dataset, the AGM module yielded consistent and significant performance gains for nearly all baseline models. For instance, Ours (YOLOv6s-AGM) achieved a relative improvement of 3.1% in mAP@50 (from 0.854 to 0.885), whereas Ours (YOLOv9c-AGM) exhibited a substantial increase in recall, rising from 0.824 to 0.865.

In the highly challenging medical task of lung nodule detection (see Table 4), the superiority of the AGM module was further validated. For both general and small nodules, models integrated with AGM were comprehensively improved. Taking Ours (YOLO11n-AGM) as an example, its mAP@50 and mAP@50-95 for general nodule detection increased to 0.924 and 0.505, respectively. This result demonstrates that AGM can effectively address the challenges of a low signal-to-noise ratio and blurred target boundaries in CT images.

TABLE 6. Performance comparison between different YOLO baseline models and their variants integrated with the AGM module on NWPU VHR-10 dataset.

Models	precision	recall	mAP@50	mAP@50-95
YOLOv8n	0.849	0.845	0.866	0.500
Ours(YOLOv8n-AGM)	0.880	0.812	0.873	0.526
YOLOv10n	0.782	0.768	0.817	0.476
Ours(YOLOv10n-AGM)	0.810	0.796	0.839	0.488
YOLO11n	0.862	0.785	0.839	0.506
Ours(YOLO11n-AGM)	0.900	0.808	0.872	0.519
YOLOv6s	0.863	0.811	0.854	0.519
Ours(YOLOv6s-AGM)	0.891	0.831	0.885	0.529
YOLOv9c	0.919	0.824	0.907	0.586
Ours(YOLOv9c-AGM)	0.919	0.865	0.909	0.591

TABLE 7. Experimental results of detection performance comparison of different models on NWPUVHR-10.

Models	Precision(%)	Recall((%)	mAP@50(%)	mAP@50-95(%)
LSKNet [33]	88.05	85.37	87.74	56.35
TPH-YOLO [34]	90.52	89.79	90.46	57.32
Super YOLO [35]	91.89	90.21	90.78	59.30
AGM-YOLO (OURS)	92.54	90.75	93.04	59.91

2) COMPARISON WITH STATE-OF-THE-ART METHODS

To contextualize our performance within the broader research landscape, we compared our best-performing AGM-YOLO models against several recently published state-of-the-art (SOTA) methods on all three datasets.

As shown in Table 3, on the BCCD dataset, our AGM-YOLO significantly outperformed other specialized methods such as YOLOv5-CW and ADA-YOLO across all key metrics. This indicates that AGM-YOLO can more accurately identify different cell types in microscopic scenes with easily overlapping targets.

On the LUNA16 benchmark (see Table 5), AGM-YOLO ranked first against competitive models such as Gold-YOLO and RT-DETR, achieving the highest precision (91.4%), recall (91.7%), and mAP@50 (93.1%). This is particularly significant given the high-precision requirements of clinical applications.

Furthermore, in the remote sensing domain (see Table 7), our method also demonstrates strong competitiveness, outperforming established methods such as TPH-YOLO and Super YOLO in mAP@50 and mAP@50-95 metrics. These collective results strongly validate that AGM-YOLO is not only effective at enhancing baselines but also achieves a state-of-the-art performance level.

3) ANALYSIS OF SYNERGY AND COMPATIBILITY

On the BCCD dataset (see Table 2), the results revealed a more nuanced interaction between the AGM module and various baseline architectures. For newer architectures, such as YOLOv8n, YOLOv6s, YOLO11n, and YOLOv9c, AGM consistently yields stable performance gains. For instance, the recall of Ours (YOLOv8n-AGM) increased from 0.842 to 0.916, and its mAP@50-95 also increased from 0.584 to 0.590.However, two phenomena worthy of in-depth discussion were observed. First, although Ours (YOLOv5-AGM) maintained a comparable performance in mAP@50 (0.910 vs. 0.898), it exhibited a slight decline in the more stringent mAP@50-95 metric (0.612 vs. 0.601). Second, Ours (YOLOv10n-AGM) shows an overall performance degradation. These observations suggest that the complex feature transformations introduced by AGM may, for classic models like YOLOv5, slightly compromise localization precision while improving recall; for YOLOv10n, which is highly optimized for extreme efffciency, the additional computational load and altered feature ffow may lead to compatibility issues. This honestly reveals the boundary conditions of "plug-and-play" modules, i.e., their effectiveness depends on the synergy with the baseline architecture.

F. ABLATION STUDY: VALIDATING THE NECESSITY OF SYNERGISTIC DESIGN

To dissect the contributions of each component within the AGM module (CMHA, GEBlock, and MFR-Neck), we conducted ablation experiments on the LUNA16 dataset using YOLOv5 as the baseline (see Table 8). The experiments first established the performance of the baseline model with an mAP@50 of 0.888. Subsequently, each component was individually incorporated. The results showed that adding CMHA, MFR-Neck, or GEBlock alone yielded performance improvements of varying degrees, confirming the independent effectiveness of each component's design. The addition of MFR-Neck (+MFR-Neck) demonstrated outstanding performance in mAP@50-95 (0.576), whereas the addition of CMHA (+CMHA) improved mAP@50 to 0.913. The most critical finding is that the final model integrating all three components, Ours (AGM-YOLOv5), achieved performance in recall (0.925) and mAP@50 (0.938), which far surpassed any single component or partial combination. This clearly demonstrates a synergistic effect (1+1+1 > 3) among CMHA's global awareness, GE's channel refinement, and MFR-Neck's multiscale demonstrating its superior sensitivity and reliability in challenging medical imaging scenarios.

TABLE 8. Comparison of detection performance when integrating different attention modules with YOLOv5 as the baseline on the LUNA16 dataset.

Models	precision	recall	mAP@50	mAP@50-95	epochs
Baseline	0.862	0.864	0.888	0.468	300
+CMHA	0.863	0.821	0.913	0.556	300
+MFR-Neck	0.884	0.8	0.919	0.576	300
+GE	0.889	0.837	0.893	0.571	300
Ours (AGM-YOLOv5)	0.869	0.925	0.938	0.583	300

G. ANALYSIS OF VISUALIZATION RESULTS

The quantitative improvements were intuitively validated by the visualization of detection results. Fig. 2 provides a qualitative comparison on a challenging scene from the BCCD dataset. It clearly shows that while the baseline YOLO11n and CAF-YOLO miss several small and overlapping platelets, our proposed method (OURS) successfully detects them. Further evidence is provided by the confusion matrices in Fig. 3. Compared to the baseline, our model's matrix [36]

shows higher diagonal values and lower off-diagonal values, indicating a more accurate classification of different blood cell categories and reduced inter-class confusion. In the CT images of LUNA16(Fig. 4), the baseline YOLOv10n exhibits low detection conffdence (e.g., 0.3) for certain nodules and even misses some, whereas OURS (YOLOv10n-AGM) can stably detect these nodules with higher conffdence (e.g., 0.9), regardless of their size. Similarly, in the complex remote sensing scenes of NWPU VHR-10 (Fig. 5), OURS (YOLOv5-AGM) successfully recalls the dense small aircraft missed by the baseline model, while OURS (YOLOv9c-AGM) corrects the misclassiffcation of the baseline model, accurately distinguishing between baseball ffelds and storage tanks.These visual evidences collectively conffrm the comprehensive capabilities of the AGM module in improving recall, enhancing localization precision, and optimizing classiffcation accuracy.

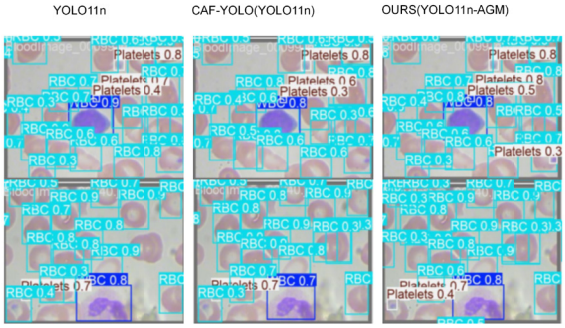

FIGURE 2. Qualitative comparison of detection performance on the BCCD dataset.

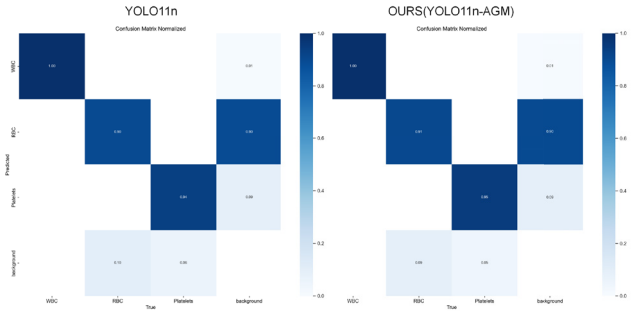

FIGURE 3. Visualization of BCCD dataset.

H. COMPREHENSIVE DISCUSSION

By synthesizing all the experimental results, the effectiveness of the proposed AGM module as a plug-and-play enhancement component was fully validated. It can significantly improve the performance of most mainstream YOLO detectors when handling small targets, particularly in challenging domains such as biomedicine and remote sensing. Ablation experiments clearly revealed that the synergistic interaction between its internal multi-attention and feature fusion mechanisms is the key to performance improvement.

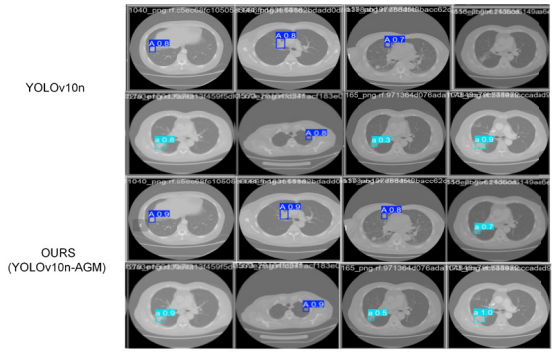

FIGURE 4. Qualitative comparison results on the LUNA16 dataset. The top two rows show the detection results of the baseline model YOLOv10n, which exhibits low detection confidence (e.g., 0.3) for certain nodules and even misses some targets (as shown in the rightmost image of the first row). In contrast, the bottom two rows display the detection results of our proposed AGM-YOLOv10n model, which can robustly detect all nodules with significantly higher detection confidence (e.g., 0.9). In this visualization, the labels "A" (dark blue, uppercase) and "a" (light blue, lowercase) are used to distinguish between large and small lung nodules, respectively. This labeling, combined with the confidence scores, enables a clear performance comparison across different nodule sizes.

FIGURE 5. Visualization of NWPU VHR-10 dataset.

The experiments also revealed the boundaries and challenges of modular design. Deep compatibility with specific baseline models is a crucial factor determining the success of "plug-and-play." Future research can be carried out in two directions: first, designing a more adaptive version of AGM that can dynamically adjust its internal strategies according to the characteristics of different baseline models; second, exploring deeper integration methods rather than merely serving as an additional module in the neck to achieve better end-to-end performance.

V. CONCLUSION

In this study, we address the formidable challenge of minute object detection in biomedical imaging and other domains. To this end, we designed and implemented an innovative plug-and-play module called AGM-YOLO. The core of this module lies in its synergistic attention architecture, which organically integrates our custom Convolutional Multi-head Self-Attention (CMHA) to capture the global context, lightweight GEBlock for channel-wise feature refinement, and efficient Multiscale Feature Refinement

Neck (MFR-Neck). This holistic co–design–rather than any single component-constitutes the primary contribution of this study through the coordinated interplay of global context capture (CMHA), channel-wise optimization (GEBlock), and multiscale feature fusion (MFR-Neck), AGM-YOLO systematically enhanced feature perception and discrimination capabilities of existing YOLO detection frameworks for minute targets, enabling a multi-dimensional and collaborative enhancement of features that single-component designs cannot achieve.

To validate its efficacy, we conducted comprehensive comparative experiments and ablation studies using three public datasets with distinct characteristics: BCCD, LUNA16, and NWPU VHR-10. The experimental results robustly demonstrate that in most scenarios, especially when tackling challenging scenes with complex backgrounds and diverse targets, such as LUNA16 and NWPU VHR-10, the AGM module significantly improves the performance of various YOLO baselines, achieving excellent results in recall and mAP. The ablation study further confirmed that the synergistic effect among the internal components of the AGM is key to maximizing performance. Concurrently, this study also candidly reveals the boundary conditions of a "plug-and-play" module: performance enhancement is not guaranteed when integrated with certain highly optimized lightweight models (e.g., YOLOv10n) or specific datasets, highlighting the critical importance of deep compatibility and co-optimization between the module and the baseline architecture.

Despite its demonstrated potential, we acknowledge that AGM-YOLO has some limitations. First, its computational complexity, primarily from the CMHA, poses challenges for its adaptation to lightweight models that prioritize extreme efficiency. Second, the model's performance exhibits a degree of dependency on the baseline architecture, which tempers its universal "plug-and-play" nature. Furthermore, like all supervised learning methods, its ability to detect rare lesions is constrained by the availability of training data, and its "black-box" nature requires improved interpretability for clinical trust.

These findings suggest several clear directions for future research. First, exploring more lightweight mechanisms for global context modeling, such as sparse attention, could reduce the computational overhead of the AGM. Second, developing an adaptive integration strategy [37] that allows the AGM module to dynamically adjust the behavior of its internal components based on the host architecture characteristics would be a valuable pursuit. Third, incorporating semi-supervised or self-supervised learning could leverage large numbers of unlabeled medical images to enhance the model's generalization capabilities. Finally, advancing research into model interpretability, such as generating clearer attention heatmaps, is crucial for building a bridge of trust between algorithms and clinicians, ultimately facilitating the practical application of high-precision automated detection technologies in clinical diagnosis. Our code will be made publicly available upon acceptance of the manuscript.

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